

Domain Controller and Active Directory Lab

Part 7 – Organizational Units(OU), Users, Account Global Domain Local Permissions (AGDLP)

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Technologies Used

- **Oracle VirtualBox**
- **Windows Server 2022**
- **Server Manager**
- **Active Directory Domain Services**

OBJECTIVE:

The objective of this lab is to create multiple Organizational Units (OU) for different departments with users assigned to them. Once the OUs are created, Account Global Group Domain Local Group Permission (AGDLP) procedure will be implemented, ensuring users assigned to departments have appropriate permissions and access.-

Lab Setup:

This lab continues from Part 6 where Dynamic Host Configuration Protocol (DHCP) was set up, providing a potential scope of leased IP ranges to clients that request an address from the DC.

For this lab we will create and assign users to OUs in the Active Directory. Users will have appropriate permissions based on the group they will become members of, minimizing risk of restricted permissions and access assigned to a user.

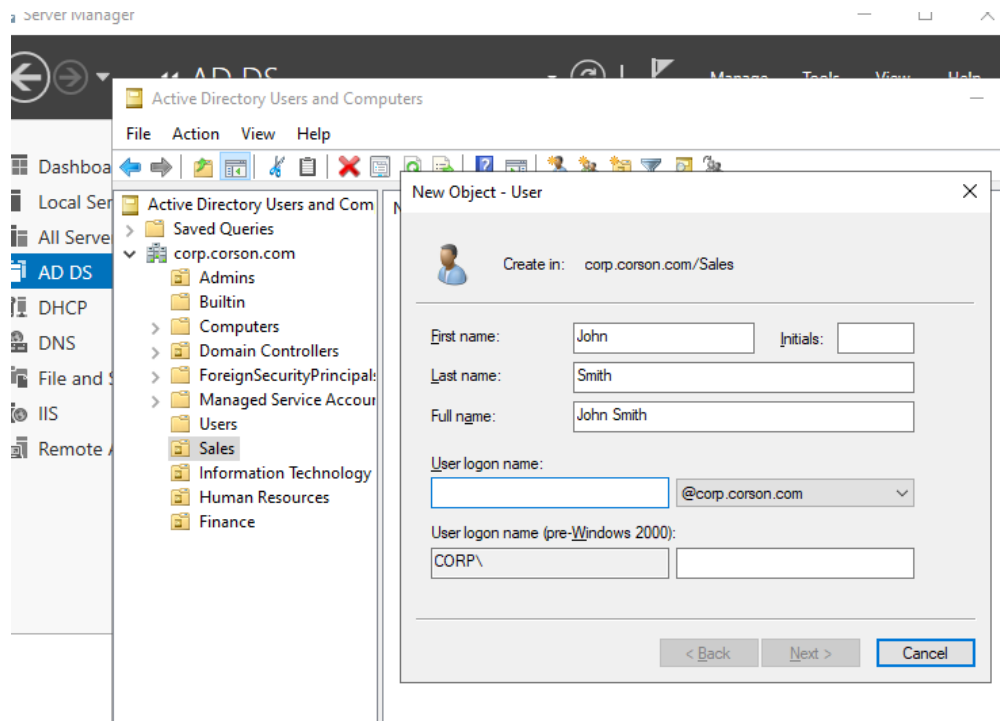


Figure 1. A new user account is created within the Sales Organizational Unit (OU), allowing the account to be organized based on the user's assigned department.

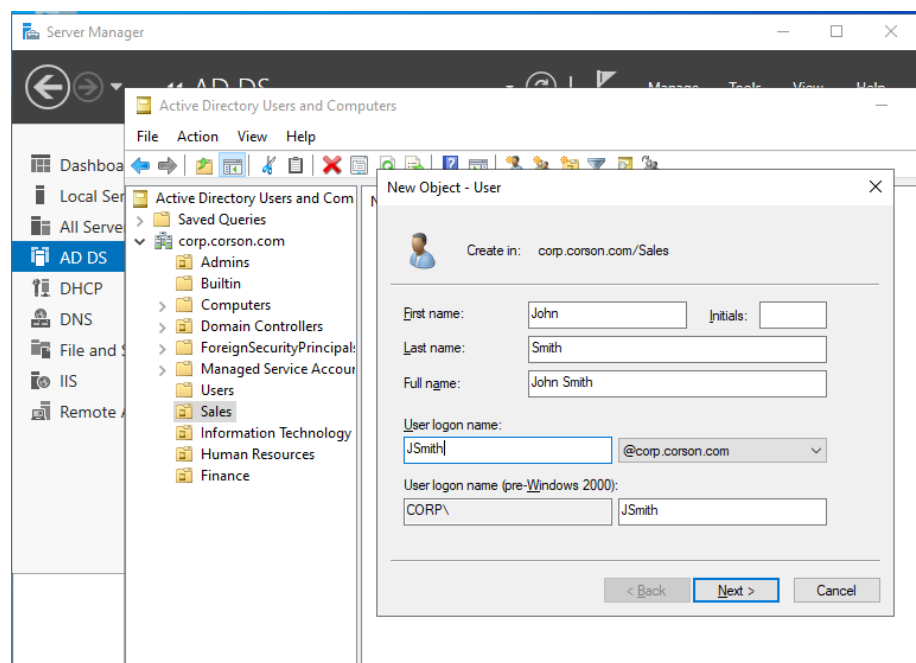


Figure 2. The user logon name JSmith assigned to John Smith following the organization's first initial and last name naming convention, providing the user with a unique account for authentication within the corp.corson.com domain.

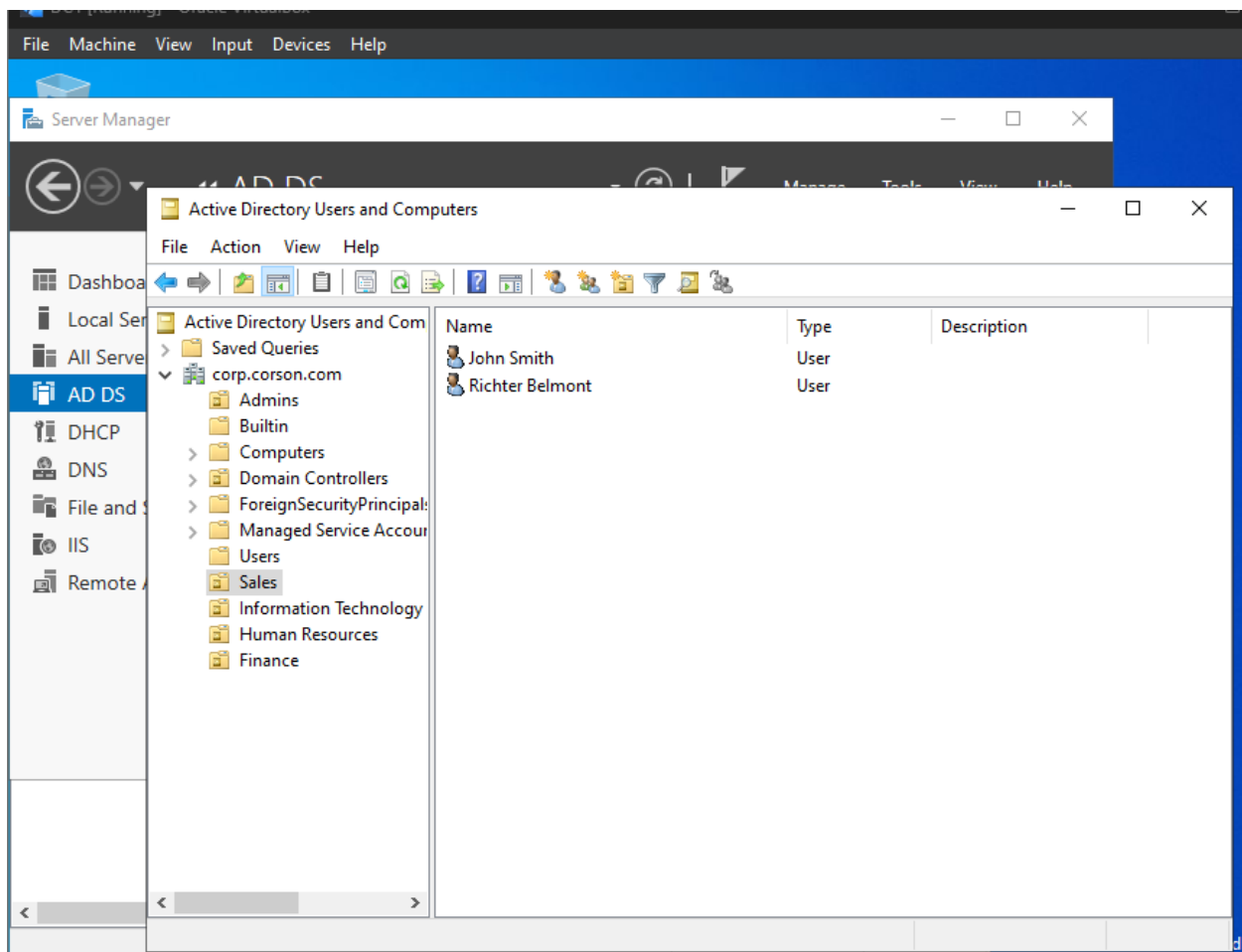


Figure 3. Multiple Organizational Units were created to represent departments within the organization, including Sales, Information Technology, Human Resources, Finance. John Smith and Richter Belmont are shown as users accounts assigned to the Sales OU.

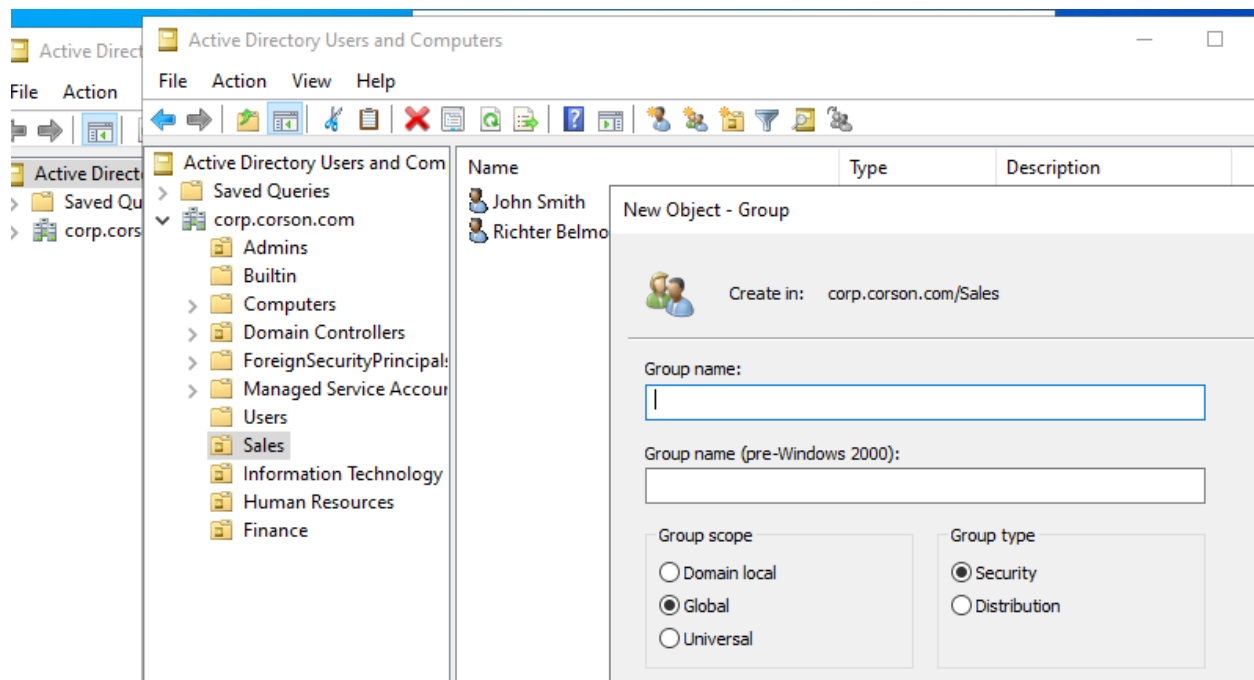


Figure 4. The GG_Sales_Users Global Security Group is created as the Global Group portion of the AGDLP model. This group provides a centralized method of grouping Sales user accounts based on their department rather than managing each account individually.

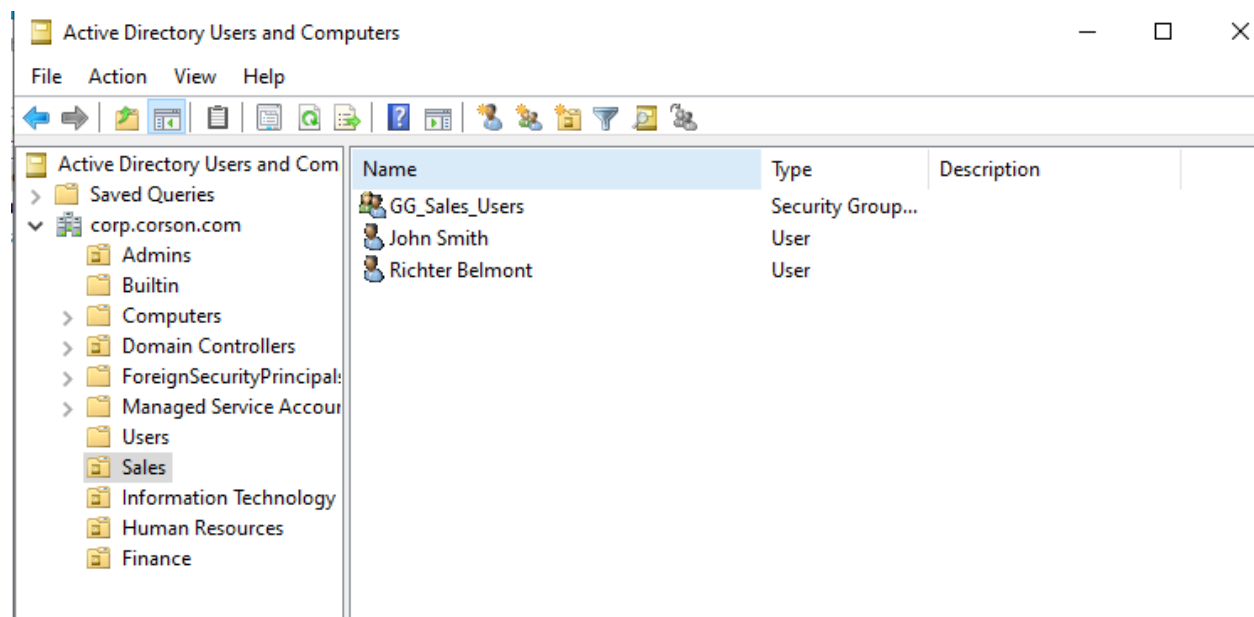


Figure 5. The GG_Sales_Users Global Security Group has been successfully created within the Sales OU and will be used to group Sales department user accounts.

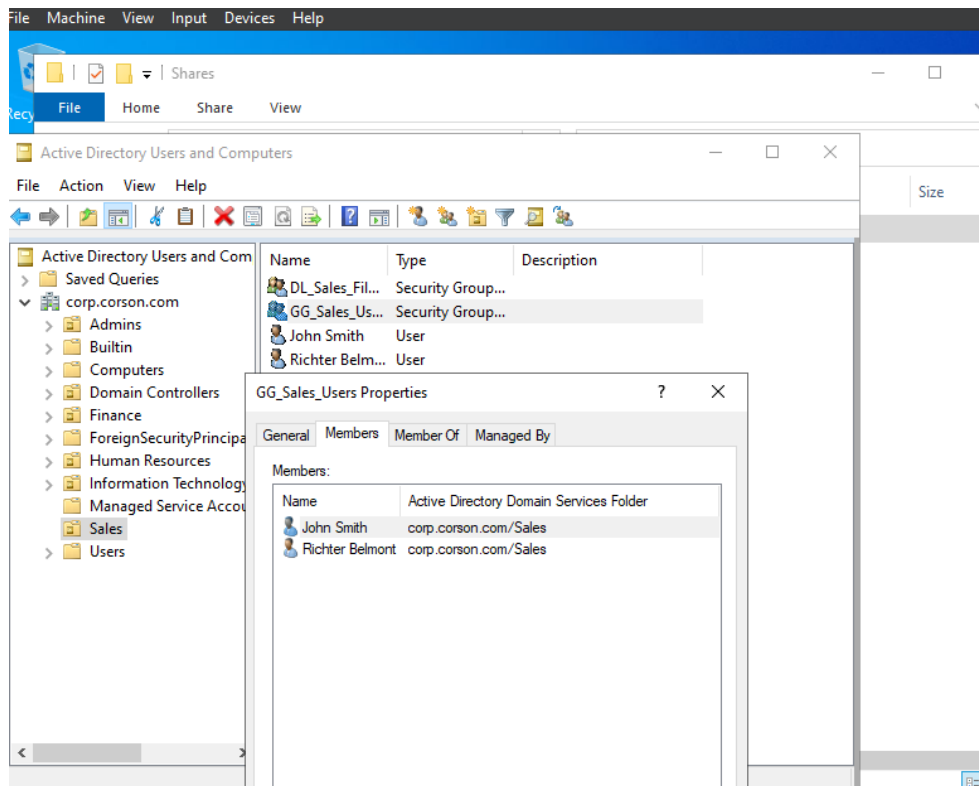


Figure 6. Both John Smith and Richter Belmont have been successfully added as members of GG_Sales_Users Global Security Group, completing the Accounts to Global portion of the AGDLP model.

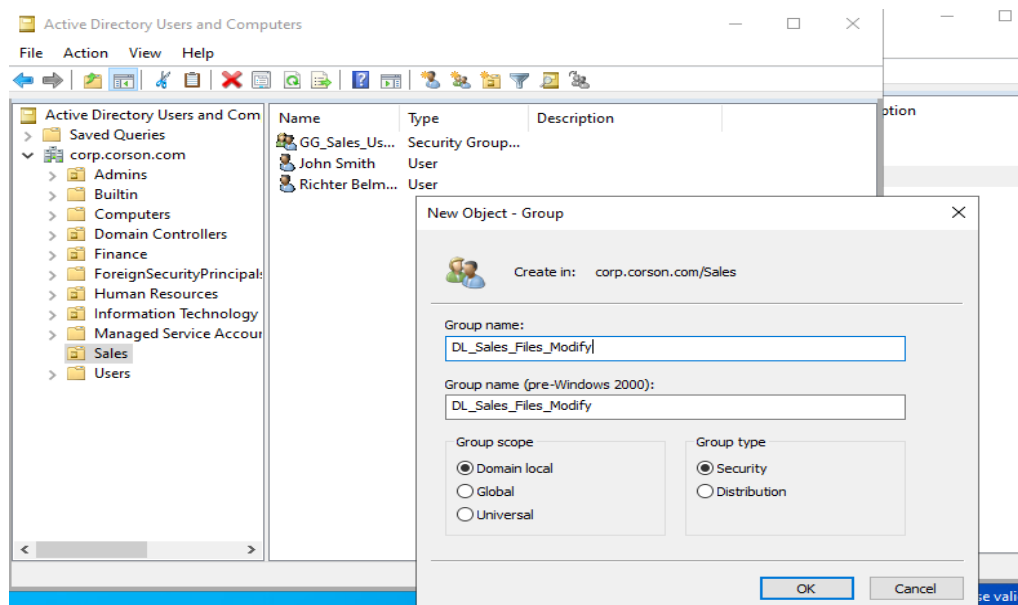


Figure 7: A Domain Local Security Group named DL_Sales_Files_Modify is created within the Sales OU. This group will be used to assign access permissions to the Sales shared folder as part of the Domain Local portion of the AGDLP model.

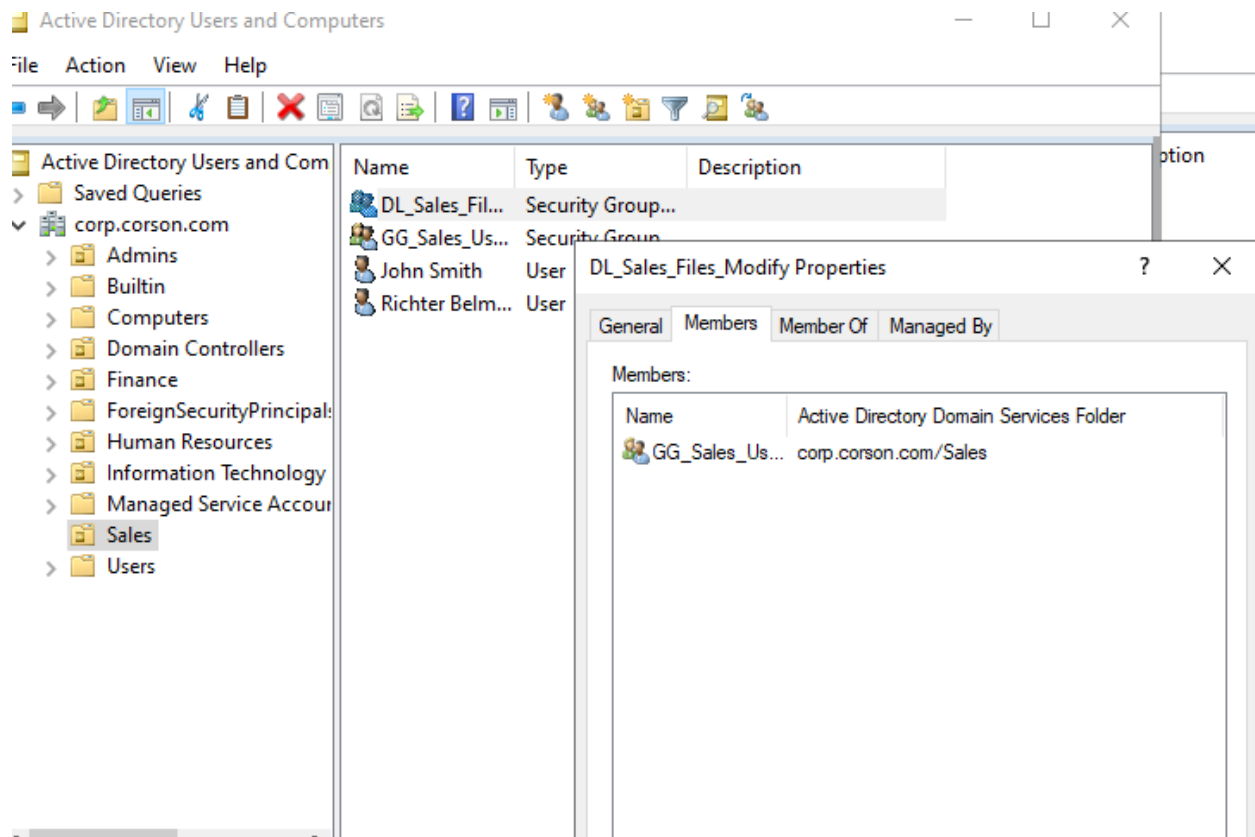


Figure 8: The GG_Sales_Users Global Security Group is added as a member of the DL_Sales_Files_Modify Domain Local Security Group. This allows the Sales users within the Global Group to receive access through the Domain Local group rather than assigning permissions individually.

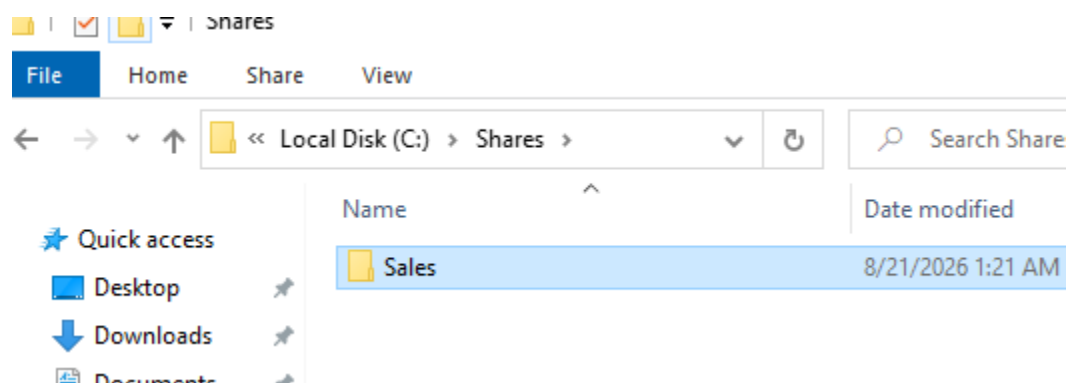


Figure 9: A Sales folder is created within the C:\Shares directory on the Domain Controller. This folder will act as a shared resource for users assigned to the Sales department.

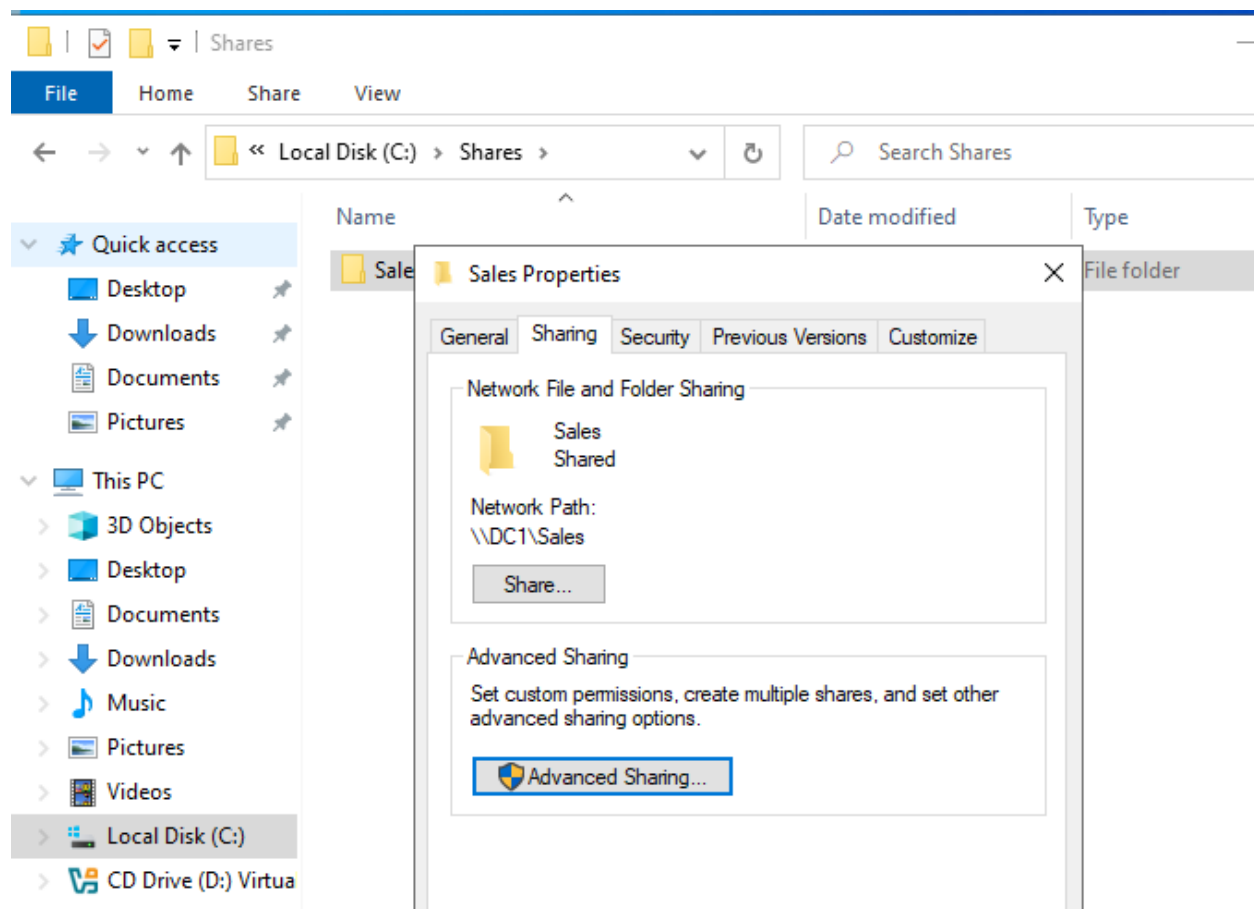


Figure 10: The Sales folder is configured as a network share through Advanced Sharing, allowing the folder to be accessed across the network using the path \\DC1\Sales.

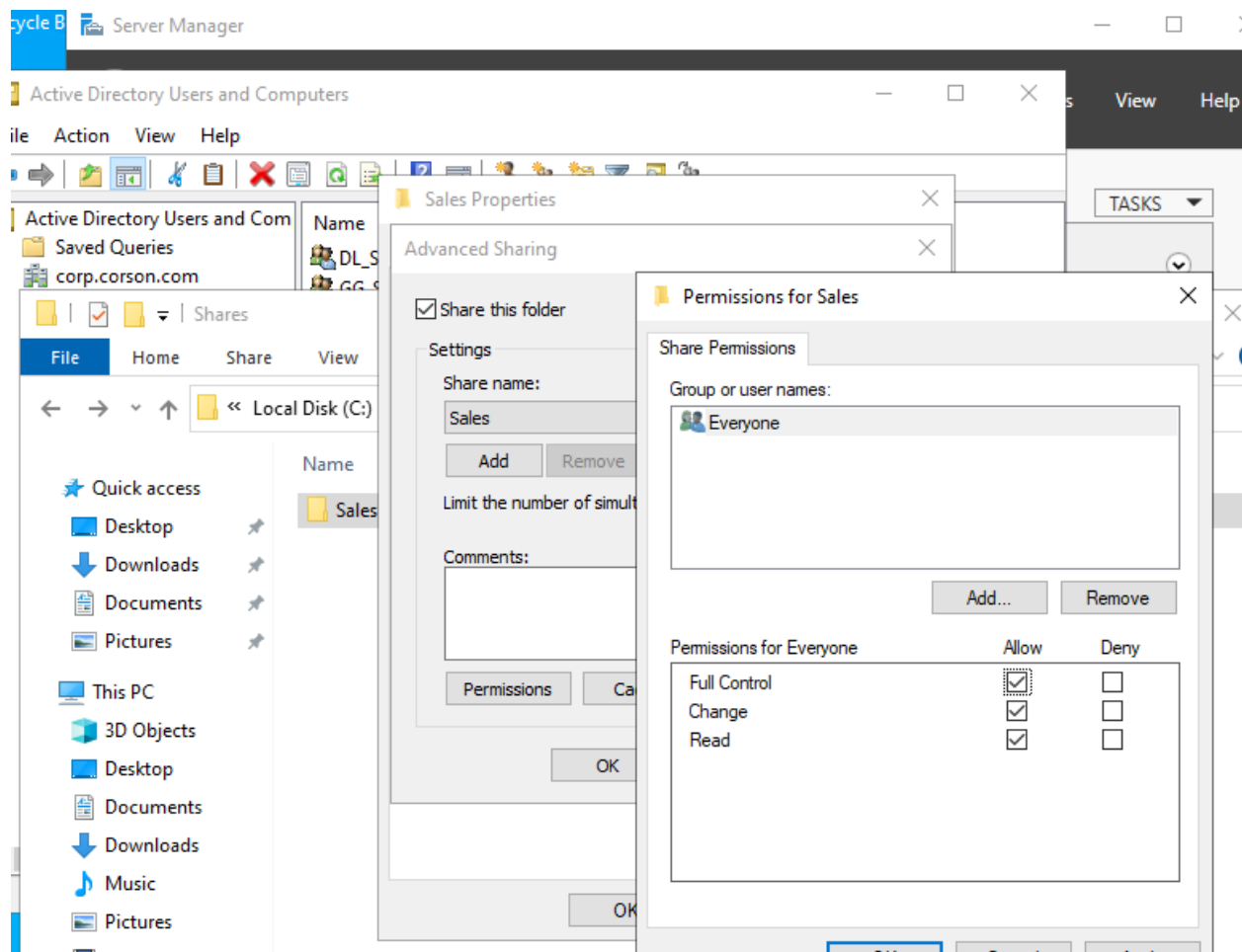


Figure 11: Share permissions are configured for the Sales network share. These permissions establish access to the shared folder over the network, while the Domain Local Security Group can be used with the folder's security permissions to control the level of access provided to Sales users.

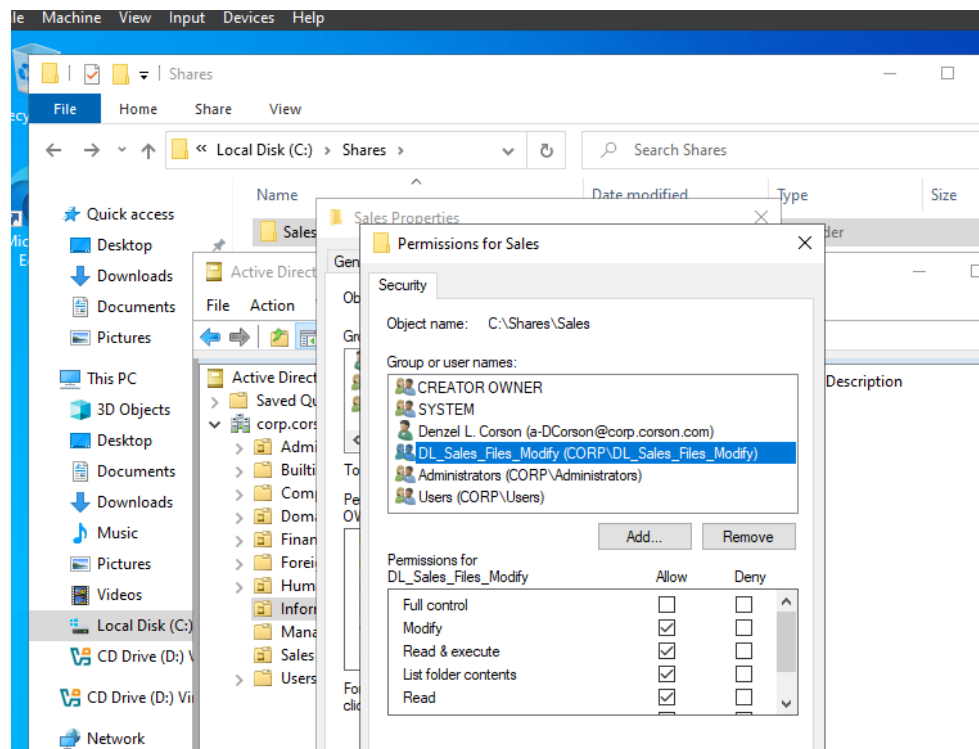


Figure 12: The DL_Sales_Files_Modify Domain Local Security Group is added to the Sales folder's NTFS permissions and assigned Modify access. This allows members of the group to read, write, modify, and delete files within the Sales folder.

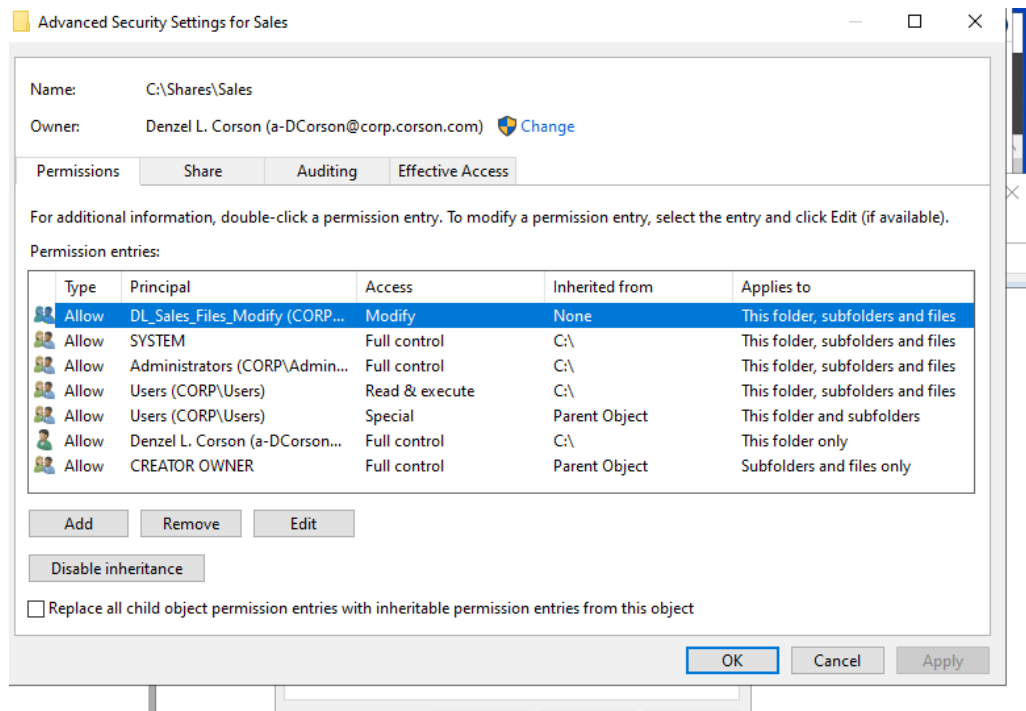


Figure 13: Advanced Security Settings verifies that DL_Sales_Files_Modify has Modify permissions applied to the Sales folder, subfolders, and files. This completes the Permissions portion of the AGDLP model

What Changed?

Before this configuration, Active Directory contained the domain and services required to support clients, but users were not organized by department or assigned access based on their roles.

Organizational Units were created for Sales, Information Technology, Human Resources, and Finance to provide a structured way of organizing users based on their assigned departments. User accounts were then created within their appropriate OU using a consistent naming convention.

The AGDLP model was introduced to manage access without assigning permissions directly to individual users. John Smith and Richter Belmont were added to the GG_Sales_Users Global Security Group, grouping the accounts based on their role within the Sales department. The GG_Sales_Users group was then added to the DL_Sales_Files_Modify Domain Local Security Group, creating the Accounts → Global Group → Domain Local Group relationship.

A Sales shared folder was also created under C:\Shares and configured as a network share. This establishes a resource that can be used to apply and test permissions for the Sales department.

This configuration provides a more organized and scalable method of managing users and access. Instead of assigning permissions to each Sales user individually, users can be added or removed from the appropriate security group as their role changes.